

Course: Fundamentals of Generative AI and Large Language Models: Theory and Practice

Week 5: Generative Adversarial Networks (GANs): Foundations

Today's Session Overview

GAN Convergence and Nash equilibrium

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GAN Convergence

The real training samples are drawn from the real-data distribution:

$$x \sim p_{data}(x)$$

The generated samples are produced as:

$$z \sim p_z(z), \quad x_{fake} = G_\phi(z)$$

The collection of generated samples follows the generator distribution:

$$x_{fake} \sim p_g(x)$$

Here:

$p_{data}(x)$ is the real-data distribution, and

$p_g(x)$ is the distribution of samples generated by G_ϕ .

At the beginning of training the generator parameters ϕ are not well trained. Therefore:

$$G_\phi(z)$$

may produce unrealistic or meaningless samples.

Hence, the generated distribution is different from the real-data distribution:

$$p_g(x) \neq p_{data}(x)$$

For a face dataset:

- real samples $x \sim p_{data}(x)$ look like human faces;
- generated samples $G_\phi(z) \sim p_g(x)$ may look like noise or distorted faces.

Therefore, the discriminator can distinguish them easily:

$$D_\theta(x) \approx 1$$

and

$$D_\theta(G_\phi(z)) \approx 0.$$

At this stage, the discriminator performs well, while the generator performs poorly.

GAN Convergence

As training progresses, the discriminator provides gradients that are backpropagated through D_θ to the generator.

The generator updates its parameters ϕ so that:

produces more realistic samples.

As the generated samples improve, the generated distribution moves closer to the real-data distribution:

$$\underline{p_g(x)} \rightarrow \underline{p_{data}(x)}$$

At the sample level, this means generated samples become harder to distinguish from real samples.

As the generator improves:

$$D_\theta(G_\phi(z)) \checkmark$$

begins to increase.

For example:

may later become:

and then:

$$D_\theta(G_\phi(z)) = 0.1 \checkmark$$

$$D_\theta(G_\phi(z)) = 0.4 \checkmark$$

$$D_\theta(G_\phi(z)) = 0.5 \checkmark$$

This means the generated images are becoming more difficult for the discriminator to classify as fake.

At ideal convergence

At the ideal GAN equilibrium:

$$\underline{p_g(x)} = \underline{p_{data}(x)}$$

The discriminator can no longer distinguish whether a sample came from the real-data distribution or the generator distribution.

Therefore:

for samples from either distribution.

In particular:

and

$$D_{\theta}(x) = \frac{1}{2}$$

$$D_{\theta}(x) \approx 0.5$$

$$D_{\theta}(G_{\phi}(z)) \approx 0.5$$

This does not mean that the discriminator is poorly trained. It means that the generator has learned to produce samples whose distribution matches the real-data distribution.

GAN convergence refers to the ideal state in which the generated-data distribution matches the real-data distribution, and the discriminator can no longer distinguish real samples from generated samples.

Nash Equilibrium in GANs

In GANs, the two players are:

Generator G and Discriminator D

At equilibrium:

and:

$$p_g(x) = p_{data}(x) ✓$$

$$D(x) = 0.5 ✓$$

- The discriminator cannot improve because real and generated samples follow the same distribution.
- The generator cannot improve further because it has already matched the real-data distribution.

Nash equilibrium is a state in which neither the generator nor the discriminator can obtain a better outcome by changing its own parameters alone, while the parameters of the other network are kept unchanged.

Next Session

Introduction to DCGAN

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Thank you